Prototype of Joint CKM and 3D Environment Reconstruction System via UAV RF Measurements

Chaoyue Zhang*, Zhiwen Zhou*, Di Wu*†, Shiqi Zeng*, Xiaoli Xu*,
Yong Zeng*†, Zaichen Zhang*† and Yongming Huang*†
*National Mobile Communications Research Laboratory, Southeast University, Nanjing 210096, China
†Purple Mountain Laboratories, Nanjing 211111, China
Email: {chaoyue_zhang, zhiwen_zhou, studywudi, 220210935, xiaolixu, yong_zeng, zczhang, huangym}@seu.edu.cn

***Abstract*—With the rapid development of low-altitude economy (LAE) and the wide application of cellular-connected unmanned aerial vehicles (UAVs), the three-dimensional (3D) radio frequency (RF) measurements have become readily available as a byproduct of communication. Learning from such historical RF measurements naturally forms a 3D channel knowledge map (CKM), which not only enhances communication performance for subsequent UAV flights, but also supports real-time environment awareness, such as 3D environment reconstruction. To validate and advance this concept, this paper introduces our developed cost-effective UAV-based CKM prototype system. By integrating onboard computing units and 5G user equipment (UE) modules on a UAV, the system achieves fully autonomous largescale 3D channel knowledge acquisition, with all data uploaded to the cloud server in real time. Leveraging the generated CKM and aided by prior knowledge of the 2D satellite map and base station (BS) locations, a 3D environmental map is reconstructed. This framework holds strong potential for future applications in 6G networks, smart cities, and LAE management.**

I. Introduction

Electric vertical take-off and landing (eVTOL) aircraft, especially unmanned aerial vehicles (UAVs) have been widely applied in multiple fields such as transportation, surveying and mapping, and future communication scenarios [1], [2]. With the fast development of the low-altitude economy (LAE), urban applications of UAVs are expanding at an unprecedented pace [3]. To support these applications, cellular-connected UAVs have been widely deployed [4]. Crucially, the mobile communication terminals onboard these UAVs can natively capture and provide partial wireless channel information. As the scale of cities grows and the number of UAVs surges, the importance of low-altitude route planning and management has become increasingly prominent. Significant research efforts have been devoted to the trajectory optimization for UAVs [5]–[7]. To ensure uninterrupted communication for UAVs, prior channel information in space is of prominent importance for UAV trajectory optimization.

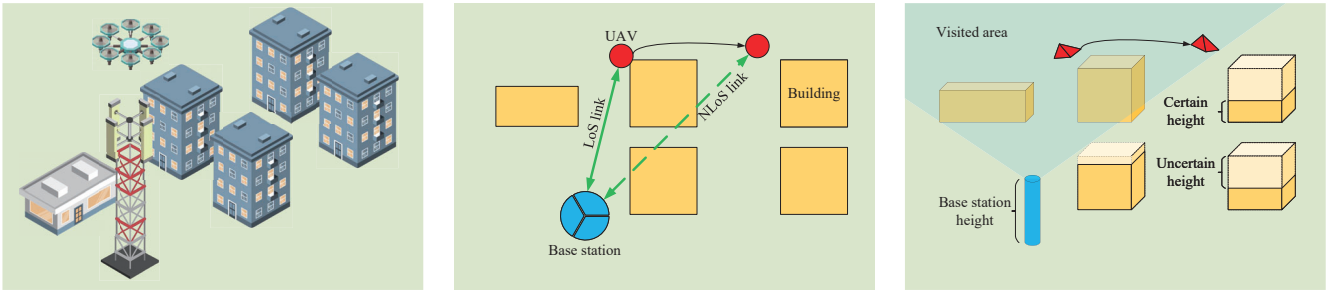
Therefore, the construction and application of channel knowledge map (CKM) play a significant role in the future development of LAE [8]. CKM is a site-specific database that provides location-specific channel prior knowledge to enhance environment-awareness [9]. Moreover, [10] studied how much data is required for CKM construction. It is important to note that access to three-dimensional (3D) environment maps is a critical prerequisite for UAV path planning. However, acquiring precise 3D environment maps remains challenging, often requiring expensive professional equipment [11]. There are open-source projects like OpenStreetMap [12], yet they face limitations including insufficient satellite measurement accuracy and poor data timeliness.

3D environment reconstruction can be achieved cost-effectively using wireless measurements [13], [14]. There has been a series of related experimental works based on the construction and application of CKM. [15] proposes an RGB-D sensor-aided ray-tracing simulation framework for indoor radio map construction that models indoor information, and indoor experiments using wireless LANs over 5180/2452 MHz clarified the accuracy of constructed radio map. [16] presented an integrated sensing and communication (ISAC)-based CKM prototype system, demonstrating the mutualistic relationship between ISAC and CKM. The prototype showcases the complete process of the construction and application of the ISAC-based CKM.

Building upon the above foundation, in this paper, we develop an autonomous prototype system to construct the 3D environment map and CKM simultaneously based on the limited radio frequency (RF) measurements along the UAV trajectory, by exploiting the inherent correlation between physical environment and wireless channel properties. As shown in Fig. 1, by equipping a UAV with a mobile communication terminal module and flying it in a specified 3D environment, the local 3D environment around commercial BS can be reconstructed in real time. Specifically, the UAV records the connection information of its serving BS and neighbor cells during the flight, and packs it with the current coordinates before uploading the data to the CKM server. Based on the measurement RF signal strength, the server determines the LoS state between the UAV and each BS, by a hypothesis test, as shown in Fig. 1b. Furthermore, the test results are used to update the height probability density function, assisted by a known 2D satellite map, as illustrated in Fig. 1c. As the measurement data increases, the height estimation becomes more accurate, and finally, 3D environment reconstruction is achieved.

II. System model

Assume that the area of interest is a cube with size $[0, D_X] \times [0, D_Y] \times [0, D_Z]$, and the point of interest within this area is



(a) Actual environment. (b) CKM construction and link state estimation. (c) Reconstructed 3D environment.

Fig. 1: Scenario of UAV measurement and CKM-based 3D environment reconstruction.

denoted by $\mathbf{q} = [q_x; q_y; q_z] \in \mathbb{R}^{3\times 1}$. Based on readily available information such as satellite maps, the outline of the occupied area of buildings in the area of interest can be obtained. We may uniformly divide the area into square grids with a side length of ΔD. Let N_{BD} be the total number of buildings in the area, and for $n = 1, \ldots, N_{\mathrm{BD}}$, let $\mathcal{B}_n$ be the set of grids contained within the boundary of the nth building. Then, the 2D physical map is denoted by:

$$\mathcal{M} \triangleq \bigcup_{n=1}^{N_{\mathrm{BD}}} \mathcal{B}_n. \tag{1}$$

There are N_{BS} BSs (indexed uniquely by i) in the area, and their locations are represented by $\mathcal{Q}^{\mathrm{BS}} = \{\mathbf{q}_i^{\mathrm{BS}} \mid i = 1, \ldots, N_{\mathrm{BS}}\}$, in which $\mathbf{q}_i^{\mathrm{BS}} = [q_{x,i}^{\mathrm{BS}}; q_{y,i}^{\mathrm{BS}}; q_{z,i}^{\mathrm{BS}}]$. Let a UAV equipped with a mobile communication user equipment (UE) module fly in this area along a 3D trajectory. During this process, the UAV may collect the BS connection information at each position $\mathbf{q}$ to form the CKM, and a total of K measurements were conducted during the task. All the measurement positions are recorded as $\mathcal{Q} = \{\mathbf{q}[k], k = 1, \ldots, K\}$. Taking the reference signal received power (RSRP) as an example, we may construct a channel gain map:

$$\mathcal{G} = \{\, g_i[k] \mid i = 1, \ldots, N_{\mathrm{BS}},\ k = 1, \ldots, K \,\}, \tag{2}$$

where $g_i[k]$ is the measured RSRP of ith BS at UAV location $\mathbf{q}[k]$. Additionally, $\mathbf{q}^{\mathrm{BS}}$ can only be obtained through measurement or query, the specific methods will be elaborated in Section IV.

We adopt the $\alpha\beta$-path loss model, for which the kth channel gain between ith BS in dB measured by the UAV is given by

$$g_i[k] = \beta_S - 10\alpha_S \log_{10} d^{\mathrm{3D}}(\mathbf{q}[k], \mathbf{q}_i^{\mathrm{BS}}) + \xi_S, \tag{3}$$

where $d^{\mathrm{3D}}(\mathbf{q}_1, \mathbf{q}_2) = ||\mathbf{q}_1 - \mathbf{q}_2||_2$ is the Euclidean distance between points $\mathbf{q}_1$ and $\mathbf{q}_2$. The link state is denoted by $S \in \{\mathrm{LoS}, \mathrm{NLoS}\}$, α_S represents the path loss exponent when the link is in state S, and β_S is the average channel gain at the reference position. ξ_S reflects the impact of shadowing effect, which follows a Gaussian distribution $\xi_S \sim \mathcal{N}(0, \sigma_S^2)$.

For a measurement point $\mathbf{q}[k]$, let $\mathcal{L}_i[k]$ represent the set of grids in the line segment between the horizontal projection $\mathbf{q}[k]$ and the ith BS, where the height of the BS is $q_{z,i}^{\mathrm{BS}} \le q_z[k]$. Consider a typical measurement, denote the number of intersecting grids between $\mathcal{L}_i[k]$ and $\mathcal{B}_n$ by M, and the center of the mth intersecting grid is denoted by $\mathbf{c}_m = [q_x^m; q_y^m]$. Let h_m represent the height of the grid, then after kth measurement, the intersection height $h_{i,n}^{\mathrm{Int}}$ is represented by the minimum h_m. It represents the height of the intersection point of the line connecting the measurement point and ith BS and the nth building. As illustrated in Fig. 2, the intersection height can be derived using similar triangles formed by the BS, the UAV measurement position, and the center of the building grid:

$$\begin{aligned} h_{i,n}^{\mathrm{Int}}[k] &= \min_{\{m \in \mathcal{L}_i[k] \cap \mathcal{B}_n\}} h_m \\ &= \min_{\{m \in \mathcal{L}_i[k] \cap \mathcal{B}_n\}} \left[\frac{d^{\mathrm{2D}}(\mathbf{c}_m, \mathbf{q}_i^{\mathrm{BS}})}{d^{\mathrm{2D}}(\mathbf{q}[k], \mathbf{q}_i^{\mathrm{BS}})} \times \left(q_z[k] - q_{z,i}^{\mathrm{BS}}\right) + q_{z,i}^{\mathrm{BS}} \right], \end{aligned} \tag{4}$$

where $d^{\mathrm{2D}}(\mathbf{q}_1, \mathbf{q}_2)$ is the distance between the projections of $\mathbf{q}_1$ and $\mathbf{q}_2$ on the X-Y plane.

In order to simultaneously construct the LoS state map and estimate building heights during UAV measurements, based on the obtained 2D physical map $\mathcal{M}$, channel gain map $\mathcal{G}$, and intersection heights h^{Int}, we first determine the link state through hypothesis testing, and then utilize this information to estimate the building heights. Further, for the N_{BD} buildings in the area, before conducting the subsequent steps, we assume that their heights $h_n, n \in N_{\mathrm{BD}}$ follow a Rayleigh distribution with the parameter σ_{pri}:

$$f^{\mathrm{pri}}(h) = \frac{h}{\sigma_{\mathrm{pri}}^2} \exp\left(-\frac{h^2}{2\sigma_{\mathrm{pri}}^2}\right). \tag{5}$$

III. 3D Environment Reconstruction via RF Measurements

Let h_n be the ground-truth height of the nth building, which can be estimated based on the RF measurements by the UAV. According to (3), the measured RSRP from UAV is related to the link state between BS and UAV. For the measured RSRP $g_i[k]$ with LoS link dominates, its probability distribution follows a Gaussian distribution $f_{\mathrm{LoS}}(g_i[k]) \triangleq \mathcal{N}(\chi_{\mathrm{LoS}}, \sigma_{\mathrm{LoS}}^2)$.

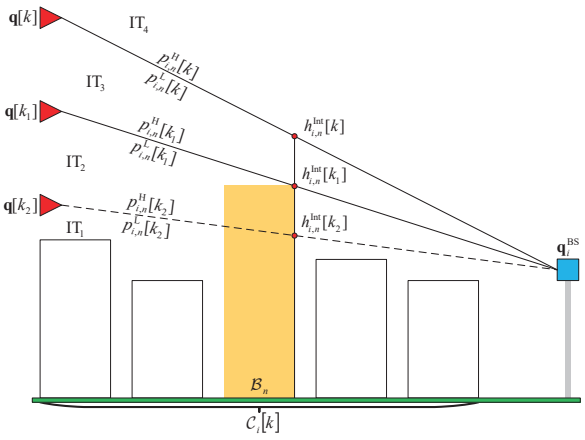


Fig. 2: The intersection height and the probability that the height of nth building is higher or lower than the intersection height.

For the case without an LoS link, the probability distribution is represented as $f_{\mathrm{NLoS}}(g_i[k]) \triangleq \mathcal{N}(\chi_{\mathrm{NLoS}}, \sigma^2_{\mathrm{NLoS}})$, where

$$\begin{aligned} \chi_{\mathrm{LoS}} &\triangleq \beta_{\mathrm{LoS}} - 10\alpha_{\mathrm{LoS}} \log_{10} d^{\mathrm{3D}}(\mathbf{q}[k], \mathbf{q}_i^{\mathrm{BS}}), \\ \chi_{\mathrm{NLoS}} &\triangleq \beta_{\mathrm{NLoS}} - 10\alpha_{\mathrm{NLoS}} \log_{10} d^{\mathrm{3D}}(\mathbf{q}[k], \mathbf{q}_i^{\mathrm{BS}}). \end{aligned} \tag{6}$$

The inference of the link state $S_i[k]$ from the measured RSRP can be formulated as a binary hypothesis testing problem:

$$\begin{cases} \mathcal{H}_0: & g_i[k] \sim \mathcal{N}(\chi_{\mathrm{LoS}}, \sigma^2_{\mathrm{LoS}}), \\ \mathcal{H}_1: & g_i[k] \sim \mathcal{N}(\chi_{\mathrm{NLoS}}, \sigma^2_{\mathrm{NLoS}}). \end{cases} \tag{7}$$

Here, $\mathcal{H}_0$ and $\mathcal{H}_1$ represent $S_i[k]$ as LoS and NLoS, respectively. The problem can be solved by:

$$\begin{cases} \mathcal{H}_0: & f_{\mathrm{LoS}}(g) > f_{\mathrm{NLoS}}(g) \\ \mathcal{H}_1: & f_{\mathrm{LoS}}(g) \le f_{\mathrm{NLoS}}(g) \end{cases} \Bigg| \; g = g_i[k]. \tag{8}$$

Let $\mathcal{C}_i[k]$ be the set of buildings that intersect with the projection of the line connecting $\mathbf{q}_i^{\mathrm{BS}}$ and $\mathbf{q}[k]$. From geometric relations, it can be deduced that if a LoS link exists between $\mathbf{q}^{\mathrm{BS}}$ and $\mathbf{q}$, the actual heights of all buildings in the intersection set should be less than or equal to the corresponding intersection height. If $S_i[k]$ is determined to be an LoS link in (8), it implies that the heights of all buildings in $\mathcal{C}_i[k]$ are less than or equal to the corresponding intersection height. That is, for all the intersecting building n, $h_n \le h_{i,n}^{\mathrm{Int}}[k]$ holds, where $h_{i,n}^{\mathrm{Int}}[k]$ can be obtained from (4). In this case, the lower height limit for all buildings in $\mathcal{C}_i[k]$ is set to $h_{i,n}^{\mathrm{Min}} = 0$, and the upper height limits are updated as:

$$h_{i,n}^{\mathrm{Max}} = \min\{h_{i,n}^{\mathrm{Max}}, h_{i,n}^{\mathrm{Int}}[k]\}, \forall n \in \mathcal{C}_i[k]. \tag{9}$$

On the other hand, if $S_i[k]$ is determined to be a NLoS link, it is considered that at least one building in $\mathcal{C}_i[k]$ has a height greater than its corresponding intersection height. LoS between the UAV and the BS may be blocked by any building with an upper limit higher than its intersection height. The set of such buildings is denoted as $\hat{\mathcal{C}}_i[k]$, in which $\exists \hat{n} \in \hat{\mathcal{C}}_i[k], h_{i,\hat{n}}^{\mathrm{Max}} > h_{i,\hat{n}}^{\mathrm{Int}}[k]$. Assuming that the probability of any building in $\hat{\mathcal{C}}_i[k]$ blocking the LoS is equal, this probability is denoted as $\bar{p}_i[k] = 1/|\hat{\mathcal{C}}_i[k]|$. That is, after the kth measurement, for each $\forall \hat{n} \in \hat{\mathcal{C}}_i[k]$, the probability is considered to be $\bar{p}_i[k]$.

As illustrated in Fig. 2, consider partitioning the z-axis of the area of interest into N_{IT} intervals with spacing h_{IT} and indexed by $j \in [1, N_{\mathrm{IT}}]$, where the lower and upper bounds of the jth interval are denoted by:

$$\begin{aligned} h^{\mathrm{L}}[j] &= h_{\mathrm{IT}} * (j-1), \\ h^{\mathrm{H}}[j] &= h_{\mathrm{IT}} * j. \end{aligned} \tag{10}$$

Whenever an NLoS link is determined, the probability distribution of the building height $f_n(h)$ is updated concurrently. Let $p_{i,n}^{\mathrm{H}}[j]$ and $p_{i,n}^{\mathrm{L}}[j]$ denote the probabilities that $h_{i,n}[k] > h^{\mathrm{H}}[j]$ and $h_{i,n}[k] < h^{\mathrm{H}}[j]$, respectively, obtained in the kth measurement. After $S_i[k]$ is determined as NLoS, the corresponding probabilities are updated based on the historical probabilities $\hat{p}_{i,n}^{\mathrm{H}}[j]$ and $\hat{p}_{i,n}^{\mathrm{L}}[j]$. At the beginning of the measurement, it is assumed that the UAV maintains LoS with all BSs, i.e., $\hat{p}_{i,n}^{\mathrm{H}}[j] = 0$ and $\hat{p}_{i,n}^{\mathrm{L}}[j] = 1$. The updates are then given by:

$$\begin{cases} p_{i,n}^{\mathrm{H}}[j] = \frac{\hat{p}_{i,n}^{\mathrm{H}}[j] + \bar{p}_i[k]}{\hat{p}_{i,n}^{\mathrm{H}}[j] + \hat{p}_{i,n}^{\mathrm{L}}[j] + \bar{p}_i[k]}, \\ p_{i,n}^{\mathrm{L}}[j] = \frac{\hat{p}_{i,n}^{\mathrm{L}}[j]}{\hat{p}_{i,n}^{\mathrm{H}}[j] + \hat{p}_{i,n}^{\mathrm{L}}[j] + \bar{p}_i[k]}. \end{cases} \tag{11}$$

Further, the normalized probability that the true building height lies in the jth interval $[h_{i,n}^{\mathrm{L}}[j], h_{i,n}^{\mathrm{H}}[j])$ is defined piecewise as:

$$p_{i,n}^{\mathrm{IT}}[j] = \begin{cases} p_{i,n}^{\mathrm{L}}[1], & j = 1 \\ p_{i,n}^{\mathrm{L}}[j] - p_{i,n}^{\mathrm{L}}[j-1], & 1 < j < N_{\mathrm{IT}} \\ p_{i,n}^{\mathrm{H}}[N_{\mathrm{IT}} - 1], & j = N_{\mathrm{IT}} \end{cases} \tag{12}$$

These interval probabilities satisfy the normalization condition $\sum_{j=1}^{N_{\mathrm{IT}}} p_{i,n}^{\mathrm{IT}}[j] = 1$. For the ith BS, the posterior probability density function of the height of the nth building is constructed as a mixture distribution:

$$f_{i,n}^{\mathrm{pos}}(h) = \sum_{j=1}^{N_{\mathrm{IT}}} p_{i,n}^{\mathrm{IT}}[j] \hat{f}_j(h), \tag{13}$$

where $\hat{f}_j(h)$ is the probability density function for the jth height interval, modeled using a truncated Rayleigh distribution:

$$\hat{f}_j(h) = \begin{cases} \dfrac{f^{\mathrm{pri}}(h)}{\exp\left(-\frac{(h_j^{\mathrm{L}})^2}{2\sigma^2_{\mathrm{pri}}}\right) - \exp\left(-\frac{(h_j^{\mathrm{H}})^2}{2\sigma^2_{\mathrm{pri}}}\right)}, & h \in [h_j^{\mathrm{L}}, h_j^{\mathrm{H}}) \\ 0, & \text{otherwise.} \end{cases} \tag{14}$$

The overall probability density function for the height of the nth building, combining information from all relevant BSs indexed by $i \in N_{\mathrm{BS}}$, is obtained by:

$$f_n(h) = \frac{\prod_{i \in N_{\mathrm{BS}}} f_{i,n}^{\mathrm{pos}}(h)}{\int_{-\infty}^{+\infty} \prod_{i \in N_{\mathrm{BS}}} f_{i,n}^{\mathrm{pos}}(h) \mathrm{d}h} \tag{15}$$

With more measurements collected, the distribution of the building heights gets more concentrated, and finally, the

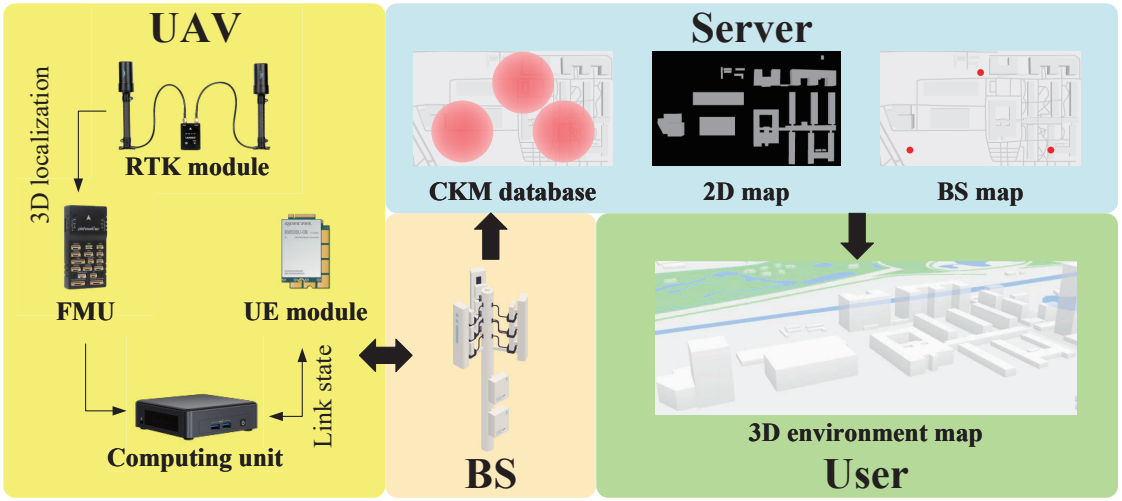


Fig. 3: System architecture of joint CKM and 3D environment reconstruction prototype.

estimated height is given by the expectation $E(h_n) = \int h f_n(h) \mathrm{d}h$, and the height of nth building is denoted by $h_n = \min\{\mathbf{h}_n^{\mathrm{Max}}, E(h_n)\}$. By updating the building heights vector $\mathbf{h} = \{h_n, n \in N_{\mathrm{BD}}\}$ within the area, efficient 3D environment reconstruction can be achieved based on RF measurements and the known 2D satellite map.

IV. Hardware and Experiment Results

A. Hardware equipment

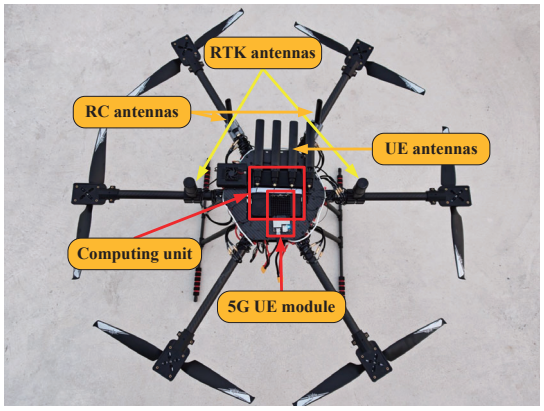


Fig. 4: The hexacopter UAV and its onboard equipment.

This section introduces the architecture of the CKM UAV prototype system, as shown in Fig. 3. The system consists of a UAV and a cloud server. The UAV maintains communication with the BS via a UE module. During flight, it continuously acquires channel state information, combines it with high-precision positioning data from a real-time kinematic (RTK) system, and generates a CKM, which is then uploaded to the cloud server. The cloud server requests 2D maps in real time, integrates known BS location data and the uploaded CKM, estimates the LoS condition of the UAV–BS link, and further constructs a 3D environment map.

UAV Platform: As illustrated in Fig. 4, the UAV is a hexacopter with a diagonal wheelbase of $850\,\mathrm{mm}$. It is equipped with a Pixhawk 6C as the flight management unit (FMU) and a UM982 RTK module for high-precision positioning. Together with an F9P RTK ground base station, the system achieves a 3D positioning accuracy of $0.8\,\mathrm{cm}$ horizontally and $1.5\,\mathrm{cm}$ vertically. A Quectel RM500U 5G UE module is used to obtain BS link information and upload data to the cloud server in real time. An ASUS NUC 15 PRO with an Intel Ultra 5 225H processor serves as the onboard computing unit, running the robot operating system 2 (ROS 2) to collect and process flight data, BS link status, and other sensor information.

Server: The server comprises a cloud database and a computing server. The database stores 2D satellite map, BS map and real-time UAV data—including 3D positions and channel measurements—which the computing server uses to construct the CKM.

B. Experiment setup

Area of interest: As shown in Fig. 5, the experiment was conducted in a $350 \times 300 \times 120\mathrm{m}^3$ area of interest, which included four major buildings.

BS map: The UE module connects to 4G-compatible BSs to obtain the location area code (LAC) and cell identity (Cell ID) required for BS location queries. Since the locations of 5G BSs are not accessible, we configure the UE module to maintain 4G connection continuously to acquire data. However, the BS locations obtained through queries may have an error of approximately 100 meters. Therefore, it is necessary to conduct secondary measurements of the BS positions using UAV horizontal and top-down views. The final measurement results are also indicated in Fig.5.

2D satellite map: To acquire the building distribution map of the area of interest, we obtain the tile map corresponding

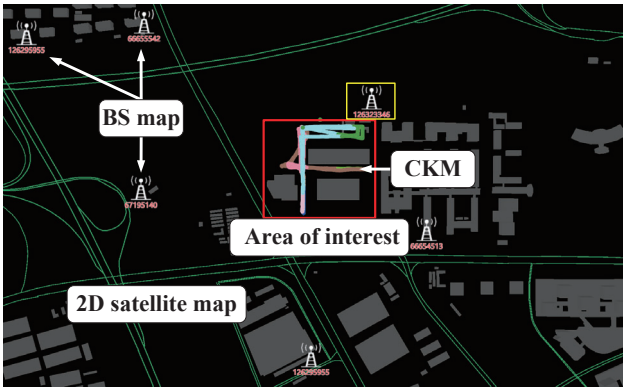


Fig. 5: Visualized data stored in the server.

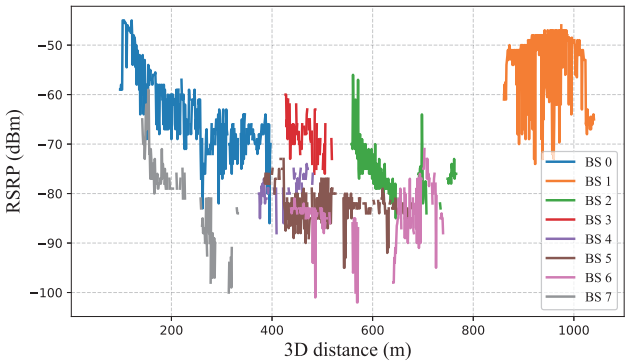


Fig. 6: RF measurement results of all BSs within the area at different 3D distances.

to the specified coordinates from the commercial map server. Subsequently, by configuring a customized map style, non-essential elements such as text labels and roads are filtered out, preserving only the building outlines. Finally, the resulting map undergoes binarization and coordinate transformation to generate the 2D meshed map required for the algorithm.

CKM: As the UAV performs 3D flight within the area, the onboard computing unit reads the link information from the UE module at discrete 3D grid points, comprising a serving cell component and a neighbor cell component. Serving Cell information refers to the BS currently serving the UE, and includes PCI, Cell ID, LAC, E-UTRA absolute radio frequency channel number (EARFCN), bandwidth, RSRP, reference signal received quality (RSRQ), received signal strength indicator (RSSI), signal-to-interference-plus-noise ratio (SINR) and other related parameters. Neighbor Cell information refers to the set of other nearby BSs detected by the UE via reference signals, from which information such as PCI, EARFCN, RSRP, RSRQ, and SINR can be obtained. It is worth noting that, due to the presence of identical PCIs within the area and the inability to obtain the neighboring cells' Cell IDs, the combination of PCI and EARFCN is considered as a unique identifier for the BSs.

C. Experiment results

Fig. 6 shows the RSRP measurements collected by the UAV from different BSs at various distances throughout the experiment process. It can be observed that due to differences in BS configurations (including transmit power and downtilt angle), the distribution of RF power in real environments is difficult to characterize with a unified path loss model, posing significant challenges for LoS state identification. Under such a complex condition without apriori information, LoS state can be estimated according to [17]. In this experiment, as an alternative, we selected some measurements from each BS that were clearly within either LoS or NLoS area, labeled their link states accordingly, and used these labeled data to train the parameters required for the hypothesis testing in (8) for subsequent processing.

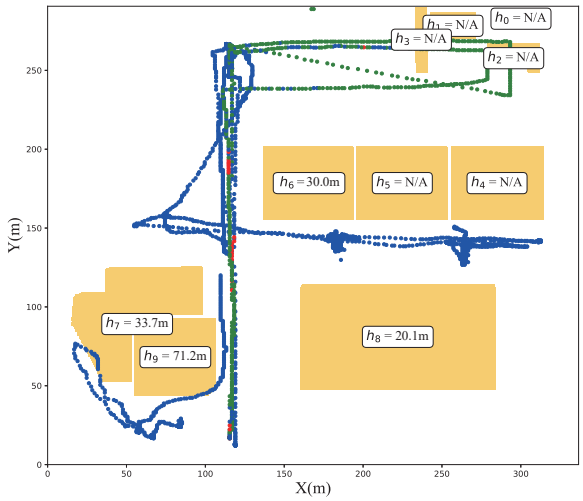


Fig. 7: Visualization of CKM (LoS state map) and meshed 2D map, along with building height estimation results.

After data preprocessing described in Section IV-B, the LoS state map and meshed building shown in Fig. 7 were obtained. For the BS marked by the yellow rectangle in Fig. 5, its LoS state map is indicated by green and red points, where the green/red points are the measurement locations that maintain LoS/NLoS with the BS, respectively. Since the 3D measurement points are projected onto a 2D plane, LoS/NLoS points may be observed simultaneously along the same flight trajectory. The remaining blue points are the other measurement points that do not contain information about this BS. Further, based on the algorithm presented in Section III, each measurement was processed individually to estimate the height of these buildings within the area, with $h_{\mathrm{IT}} = 10\,\mathrm{m}$ and $N_{\mathrm{IT}} = 12$. The estimation results indicate that the heights of buildings 6, 7, 8, and 9, where measurements are relatively concentrated, were successfully estimated, whereas no estimation results were obtained for buildings 0-5. Specifically, buildings 0-3 are low-rise structures on the ground with heights less than $10\,\mathrm{m}$, making them unestimable. For

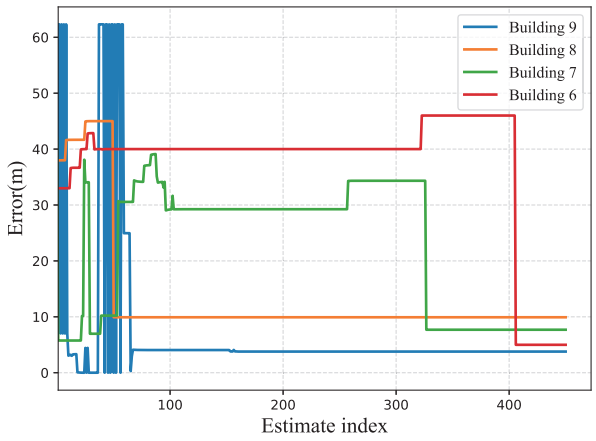


Fig. 8: The height estimation error of most buildings reduces to 10 m after collecting of 50 effective measurements. Similarly, the estimated height converge for other buildings with the collection of more measurements.

buildings 4 and 5, since the BSs are mainly located on the left side, when the UAV flew rightward to building 5, significant blockage by surrounding buildings prevented the detection of neighboring cell reference signals, resulting in insufficient data for estimation. A similar problem occurred with building 8, where the abundance of invalid data led to relatively large errors in height estimation. Finally, for buildings 6, 7, and 9, due to the relatively open environment, UAV flights were conducted around these buildings.

Fig. 8 illustrates the process of height estimation for each building by the algorithm. It can be observed that the estimation error often suddenly increases or decreases at specific moments, which is attributed to the influence of UAV flight trajectory. According to the building height estimation method described in Section III, when there is only a single obstacle between the BS and UE, the measurement at the moment when the UAV switches from LoS to NLoS has a significant impact on the estimated building height. However, during constant altitude flights, the link state typically does not change. Therefore, abrupt changes in height estimation occur during UAV takeoff and landing. The final height estimation errors are all less than h_{IT}, which meets the experimental expectations.

V. Conclusion

In this paper, we have implemented a complete prototype system for 3D CKM construction, further derived the LoS conditions between each measurement point and every BS, and finally estimated the heights of buildings within the area along with their probability distributions based on sparse measurement data. In future research, the automatic link state estimation algorithm for complex low-altitude environments will be improved; on the other hand, an AI-based sparse sampling CKM inpainting algorithm will be deployed to further enhance the application reliability of the prototype system in real-world environments. This is expected to ultimately support applications such as high-precision fingerprint-based localization for UAVs and training-free communication resource allocation.

Acknowledgment

This work was supported by the National Natural Science Foundation of China under Grant 62571116, and also by the "Program for Innovative Talents and Entrepreneur in Jiangsu" under grant number 1104000402.

References

[1] A. Baltaci, E. Dinc, M. Ozger, A. Alabbasi, C. Cavdar, and D. Schupke, "A survey of wireless networks for future aerial communications (FACOM)," *IEEE Communications Surveys & Tutorials*, vol. 23, no. 4, pp. 2833–2884, 2021.

[2] Y. Song, Y. Zeng, Y. Yang, Z. Ren, G. Cheng, X. Xu, J. Xu, S. Jin, and R. Zhang, "An overview of cellular ISAC for low-altitude UAV: New opportunities and challenges," *IEEE Communications Magazine*, pp. 1–8, 2025.

[3] H. Shakhatreh, A. H. Sawalmeh, A. Al-Fuqaha, Z. Dou, E. Almaita, I. Khalil, N. S. Othman, A. Khreishah, and M. Guizani, "Unmanned aerial vehicles (UAVs): A survey on civil applications and key research challenges," *IEEE Access*, vol. 7, pp. 48 572–48 634, 2019.

[4] Y. Zeng, Q. Wu, and R. Zhang, "Accessing from the sky: A tutorial on UAV communications for 5G and beyond," *Proceedings of the IEEE*, vol. 107, no. 12, pp. 2327–2375, 2019.

[5] Y. Zeng and R. Zhang, "Energy-efficient UAV communication with trajectory optimization," *IEEE Transactions on Wireless Communications*, vol. 16, no. 6, pp. 3747–3760, 2017.

[6] Y. Li, X. Wang, Z. Zheng, J. Guo, and Z. Fei, "Joint trajectory and transmit power design for cellular-connected UAVs via differentiable channel knowledge map," *IEEE Transactions on Vehicular Technology*, vol. 74, no. 10, pp. 15 772–15 788, 2025.

[7] T. Wang, C. Zhang, and H. Liang, "CKM-assisted UAV communication design," in *2024 IEEE 99th Vehicular Technology Conference (VTC2024-Spring)*, 2024, pp. 1–6.

[8] Y. Zeng and X. Xu, "Toward environment-aware 6G communications via channel knowledge map," *IEEE Wireless Communications*, vol. 28, no. 3, pp. 84–91, 2021.

[9] Y. Zeng, J. Chen, J. Xu, D. Wu, X. Xu, S. Jin, X. Gao, D. Gesbert, S. Cui, and R. Zhang, "A tutorial on environment-aware communications via channel knowledge map for 6G," *IEEE Communications Surveys & Tutorials*, vol. 26, no. 3, pp. 1478–1519, 2024.

[10] X. Xu and Y. Zeng, "How much data is needed for channel knowledge map construction?" *IEEE Transactions on Wireless Communications*, vol. 23, no. 10, pp. 13 011–13 021, 2024.

[11] Y. Xu, P. Ma, E. Ng, and H. Lin, "Fusion of WorldView-2 stereo and multitemporal TerraSAR-X images for building height extraction in urban areas," *IEEE Geoscience and Remote Sensing Letters*, vol. 12, no. 8, pp. 1795–1799, 2015.

[12] M. Haklay and P. Weber, "OpenStreetMap: User-generated street maps," *IEEE Pervasive Computing*, vol. 7, no. 4, pp. 12–18, 2008.

[13] A. T. Irish, J. T. Isaacs, F. Quitin, J. P. Hespanha, and U. Madhow, "Probabilistic 3D mapping based on GNSS SNR measurements," in *2014 IEEE international conference on acoustics, speech and signal processing (ICASSP)*. IEEE, 2014, pp. 2390–2394.

[14] J. Kim, J. A. Zalat, Y. Bahoo, and S. Saeedi, "Structure from WIFI (SFW): Rssi-based geometric mapping of indoor environments," in *2024 American Control Conference (ACC)*. IEEE, 2024, pp. 259–264.

[15] N. Suga, Y. Maeda, and K. Sato, "Indoor radio map construction via ray tracing with RGB-D sensor-based 3D reconstruction: Concept and experiments in WLAN systems," *IEEE Access*, vol. 11, pp. 24 863–24 874, 2023.

[16] C. Zhang, Z. Zhou, X. Xu, Y. Zeng, Z. Zhang, and S. Jin, "Prototyping and experimental results for ISAC-based channel knowledge map," *IEEE Transactions on Vehicular Technology*, vol. 74, no. 7, pp. 10 719–10 731, 2025.

[17] K. Sato, K. Suto, K. Inage, K. Adachi, and T. Fujii, "Space-frequency-interpolated radio map," *IEEE Transactions on Vehicular Technology*, vol. 70, no. 1, pp. 714–725, 2021.